

Multilinear regression as loss minimization.

Goals

- Adding regressors based on discrete random variables
 - The one-hot encoding trick
 - Sample means and one-hot encoding
 - Adding interactions to fit different slopes

Reading

These notes are supplementary for the reading

- Ding (2024) Chapter 16 (Section 17.3 also has some interesting reading)
- Gelman, Hill, and Vehtari (2021) Sections 10.3–10.4

Ames housing data

Let's consider an example of the [Ames housing data](#), restricted to normal sales condition and considering only residential sales with either good or excellent condition kitchens. I'll be particularly interested in whether there's evidence that the cost of a kitchen upgrade is reflected in the sale value of the house.

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.1      v tibble     3.2.1
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.0.2
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

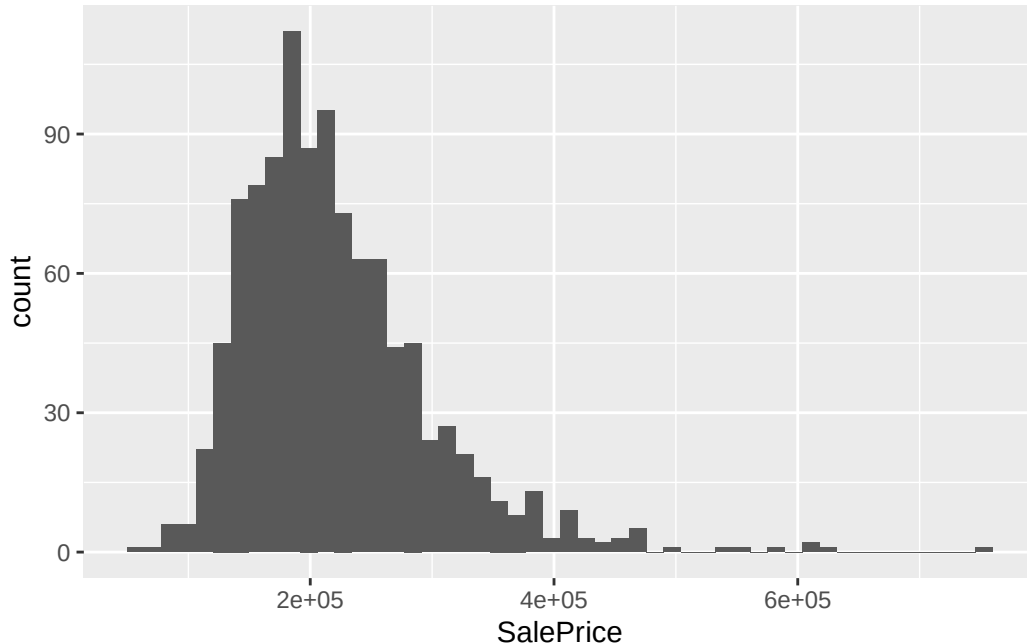
Attaching package: 'gridExtra'

The following object is masked from 'package:dplyr':

combine

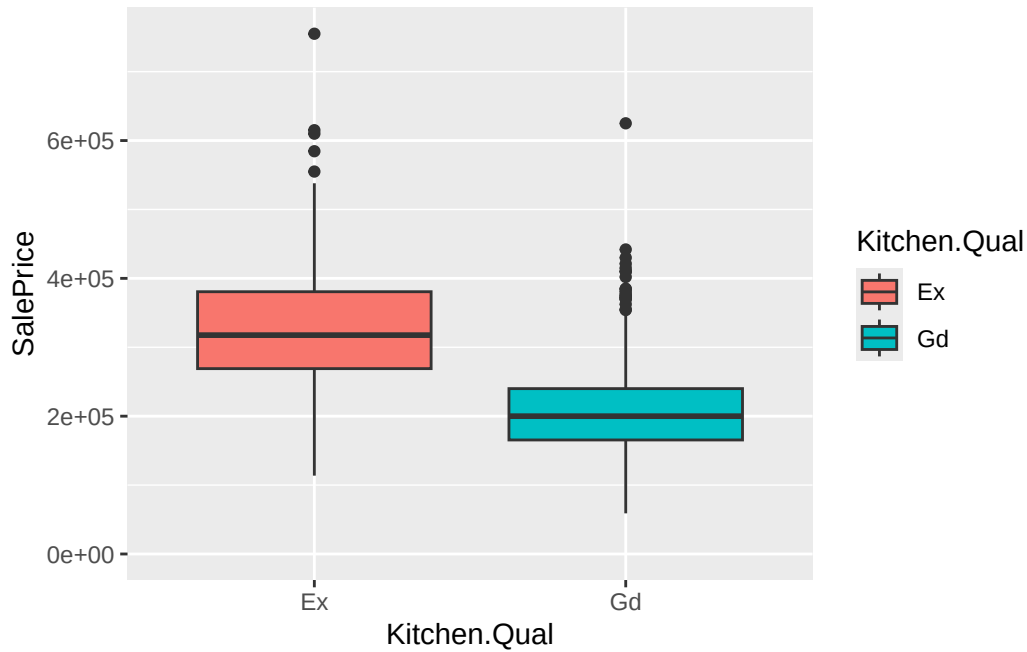
```
housing_dir <- file.path(root_dir, "datasets/ames_house/data")
ames_orig <- read.table(file.path(housing_dir, "AmesHousing.txt"), sep="\t", header=T)
ames <- ames_orig %>%
  filter(Sale.Condition == "Normal",
         # remove agricultural, commercial and industrial
         !(MS.Zoning %in% c("A (agr)", "C (all)", "I (all)"))) %>%
  filter(Kitchen.Qual %in% c("Gd", "Ex")) %>%
  mutate(Overall.Qual=factor(Overall.Qual))

ggplot(ames) +
  geom_histogram(aes(x=SalePrice), bins=50)
```

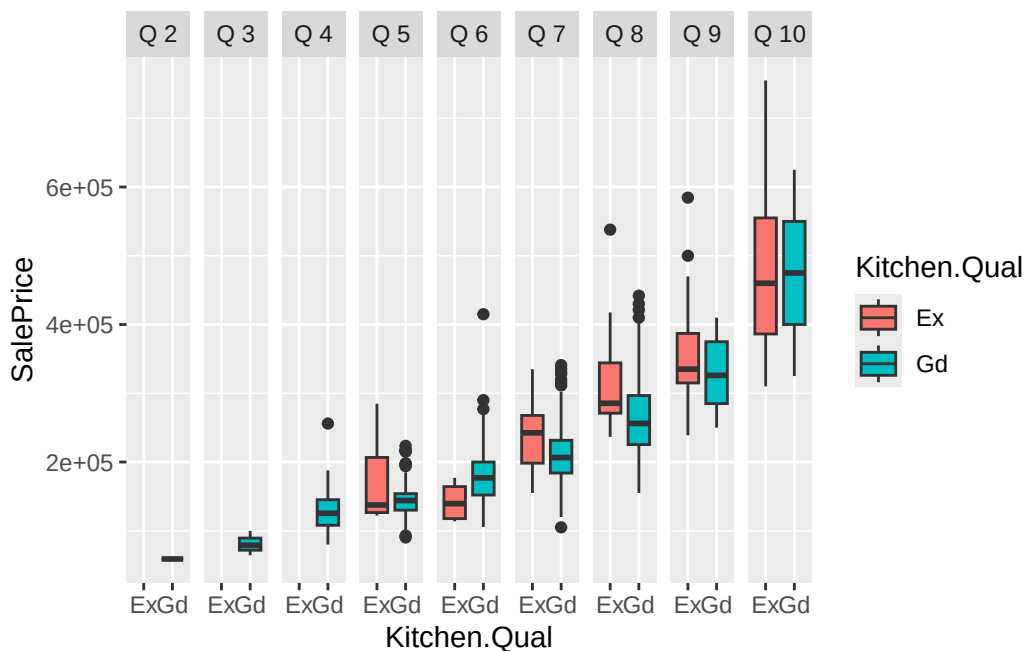


Here's a histogram of the sales price:

```
ames %>%  
  ggplot() +  
  geom_boxplot(aes(x=Kitchen.Qual, y=SalePrice, fill=Kitchen.Qual))+  
  expand_limits(y=0)
```



The average difference between Excellent and Good kitchens is \$119179, which is a huge amount of money. However, this difference mostly vanishes when we look only at differences within “Overall Quality:”



How can we study this using regression? Note that “Kitchen Quality” and “Overall Quality” are discrete variables. Here, they do have an ordering, but suppose we just want to treat them as distinct categories — in this way, we can learn how to account for other discrete variables, such as neighborhood, that are harder to visualize.

One-hot encodings

How can we use categorical variables in a regression? Suppose we have a columns $k_n \in \{g, e\}$ indicating whether a kitchen is “good” or “excellent”. A one-hot encoding of this categorical variable is given by

$$x_{ng} = \begin{cases} 1 & \text{if } k_n = g \\ 0 & \text{if } k_n \neq g \end{cases} \quad x_{ne} = \begin{cases} 1 & \text{if } k_n = e \\ 0 & \text{if } k_n \neq e \end{cases}.$$

We can then regress $y_n \sim \beta_g x_{ng} + \beta_e x_{ne} = x_n^\top \beta$. The corresponding X matrix might look like

$$k = \begin{pmatrix} g \\ e \\ g \\ g \\ \vdots \end{pmatrix} \quad X = (x_g \quad x_e) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 0 \\ \vdots & \vdots \end{pmatrix}$$

Note that $x_g^\top x_g$ is just the number of entries with $k_n = g$, and $x_g^\top x_e = 0$ because a kitchen is either good or excellent but never both.

We then have

$$X^\top X = \begin{pmatrix} x_g^\top x_g & x_g^\top x_e \\ x_e^\top x_g & x_e^\top x_e \end{pmatrix} = \begin{pmatrix} N_g & 0 \\ 0 & N_e \end{pmatrix}.$$

Then $X^\top X$ is invertible as long as $N_g > 0$ and $N_e > 0$, that is, as long as we have at least one observation of each kitchen type, and

$$(X^\top X)^{-1} = \begin{pmatrix} \frac{1}{N_g} & 0 \\ 0 & \frac{1}{N_e} \end{pmatrix}.$$

Similarly, $x_g^\top Y$ is just the sum of entries of Y where $k_n = g$, with the analogous conclusion for x_e . From this we recover the result that

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y = \begin{pmatrix} \frac{1}{N_g} & 0 \\ 0 & \frac{1}{N_e} \end{pmatrix} \begin{pmatrix} \sum_{n:k_n=g} y_n \\ \sum_{n:k_n=e} y_n \end{pmatrix} = \begin{pmatrix} \frac{1}{N_g} \sum_{n:k_n=g} y_n \\ \frac{1}{N_e} \sum_{n:k_n=e} y_n \end{pmatrix}.$$

If we let $\bar{y}_e$ and $\bar{y}_g$ denote the sample means within each group, we have shown that $\hat{\beta}_g = \bar{y}_g$ and $\hat{\beta}_e = \bar{y}_e$, as we proved before without using the matrix formulation.

Ames example

Note that we get the same thing by just computing means and running a regression in the Ames data:

```
ames %>%
  group_by(Kitchen.Qual) %>%
  summarize(mean_price=mean(SalePrice)) %>%
  pivot_wider(names_from=Kitchen.Qual, values_from=mean_price) %>%
  print()
```

```
# A tibble: 1 x 2
      Ex      Gd
  <dbl> <dbl>
1 327007. 207828.
```

```
reg <- lm(SalePrice ~ Kitchen.Qual - 1, ames)
print(coefficients(reg))
```

```
Kitchen.QualEx Kitchen.QualGd
      327006.5      207828.0
```

And when we look at the model matrix for the regression, we see the one-hot encoding in action:

```
x <- model.matrix(SalePrice ~ Kitchen.Qual - 1, ames)
bind_cols(
  select(ames, Kitchen.Qual),
  x) %>%
  head(10)
```

	Kitchen.Qual	Kitchen.QualEx	Kitchen.QualGd
1	Gd	0	1
2	Ex	1	0
3	Gd	0	1
4	Gd	0	1
5	Gd	0	1
6	Gd	0	1
7	Gd	0	1
8	Gd	0	1
9	Gd	0	1
10	Ex	1	0

What if we included a constant?

```
reg_const <- lm(SalePrice ~ Kitchen.Qual, ames)
print(coefficients(reg_const))
```

```
(Intercept) Kitchen.QualGd
      327006.5      -119178.5
```

You will show in your homework that these two regressions are equivalent, but that the interpretation of their coefficients is different. In this case, the equivalence is clearest from the prediction function:

```
reg <- lm(SalePrice ~ Kitchen.Qual - 1, ames)
reg_const <- lm(SalePrice ~ Kitchen.Qual, ames)

good_df <- data.frame(Kitchen.Qual="Gd")
ex_df <- data.frame(Kitchen.Qual="Ex")

cat("Estimated difference without regressing on a constant: ",
    predict(reg, ex_df) - predict(reg, good_df), "\n")
```

Estimated difference without regressing on a constant: 119178.5

```
cat("Estimated difference regressing on a constant: ",
    predict(reg_const, ex_df) - predict(reg_const, good_df), "\n")
```

Estimated difference regressing on a constant: 119178.5

The key is to interpret the coefficients by plugging in multiple values and seeing what the fitted model predicts. Depending on how you set up your regression, this may or may not correspond to a particular coefficient value!

Controlling for multiple variables

We can now regress on kitchen quality “controlling for” overall quality (scare quotes intentional). One way to do this is to regress on one-hot indicators for both:

```
reg_qual <- lm(SalePrice ~ Overall.Qual + Kitchen.Qual, ames)
print(coefficients(reg_qual))
```

(Intercept)	Overall.Qual3	Overall.Qual4	Overall.Qual5	Overall.Qual6
89395.19	22100.00	73468.42	86982.22	119942.62
Overall.Qual7	Overall.Qual8	Overall.Qual9	Overall.Qual10	Kitchen.QualGd
150197.56	206392.52	266350.76	392470.83	-30395.19

Note that the increase in sales price associated with an Excellent kitchen is much lower after you control for overall quality of the house in this way. **It appears that much of the difference in cost in the above regression may be attributed to correlation between kitchen quality and overall house quality.**

Note that the above regression assumes that the “effect” of overall quality is the same irrespective of the kitchen quality. We could run a regression with all the interactions to separately estimate the change in kitchen quality for each house quality. This is what actually corresponds to the graph above:

```
reg_qual <- lm(SalePrice ~ Overall.Qual * Kitchen.Qual - 1, ames)
print(summary(reg_qual))
```

Call:

```
lm(formula = SalePrice ~ Overall.Qual * Kitchen.Qual - 1, data = ames)
```

Residuals:

Min	1Q	Median	3Q	Max
-168246	-29701	-4211	22776	276754

Coefficients: (3 not defined because of singularities)

	Estimate	Std. Error	t value	Pr(> t)
Overall.Qual2	62246	59361	1.049	0.294604
Overall.Qual3	84346	45100	1.870	0.061737 .
Overall.Qual4	135715	37507	3.618	0.000311 ***
Overall.Qual5	169931	16713	10.168	< 2e-16 ***
Overall.Qual6	142425	23636	6.026	2.33e-09 ***
Overall.Qual7	239900	13646	17.580	< 2e-16 ***
Overall.Qual8	311166	9271	33.565	< 2e-16 ***
Overall.Qual9	355043	6493	54.679	< 2e-16 ***
Overall.Qual10	478246	13111	36.478	< 2e-16 ***
Kitchen.QualGd	-3246	35905	-0.090	0.927979
Overall.Qual3:Kitchen.QualGd	NA	NA	NA	NA
Overall.Qual4:Kitchen.QualGd	NA	NA	NA	NA
Overall.Qual5:Kitchen.QualGd	-20254	39849	-0.508	0.611362
Overall.Qual6:Kitchen.QualGd	41057	43112	0.952	0.341146
Overall.Qual7:Kitchen.QualGd	-27466	38491	-0.714	0.475649
Overall.Qual8:Kitchen.QualGd	-44328	37218	-1.191	0.233910
Overall.Qual9:Kitchen.QualGd	-23061	39172	-0.589	0.556178
Overall.Qual10:Kitchen.QualGd	NA	NA	NA	NA

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 47270 on 1041 degrees of freedom
Multiple R-squared: 0.9598, Adjusted R-squared: 0.9592
F-statistic: 1657 on 15 and 1041 DF, p-value: < 2.2e-16

The problem is these are extremely variable. Using additive effects can be thought of smoothing out the data so you can do “as-if” comparisons within a particular bucket.

Ding, Peng. 2024. “Linear Model and Extensions.” *arXiv Preprint arXiv:2401.00649*.

Gelman, Andrew, Jennifer Hill, and Aki Vehtari. 2021. *Regression and Other Stories*. Cambridge University Press.